Descriptive Analytics and KPI Tables

By: SEK SOCHEAT
Mobile: 017 879 967
E-mail: socheat.sek@gmail.com

WEEK 3 LECTURE DECK

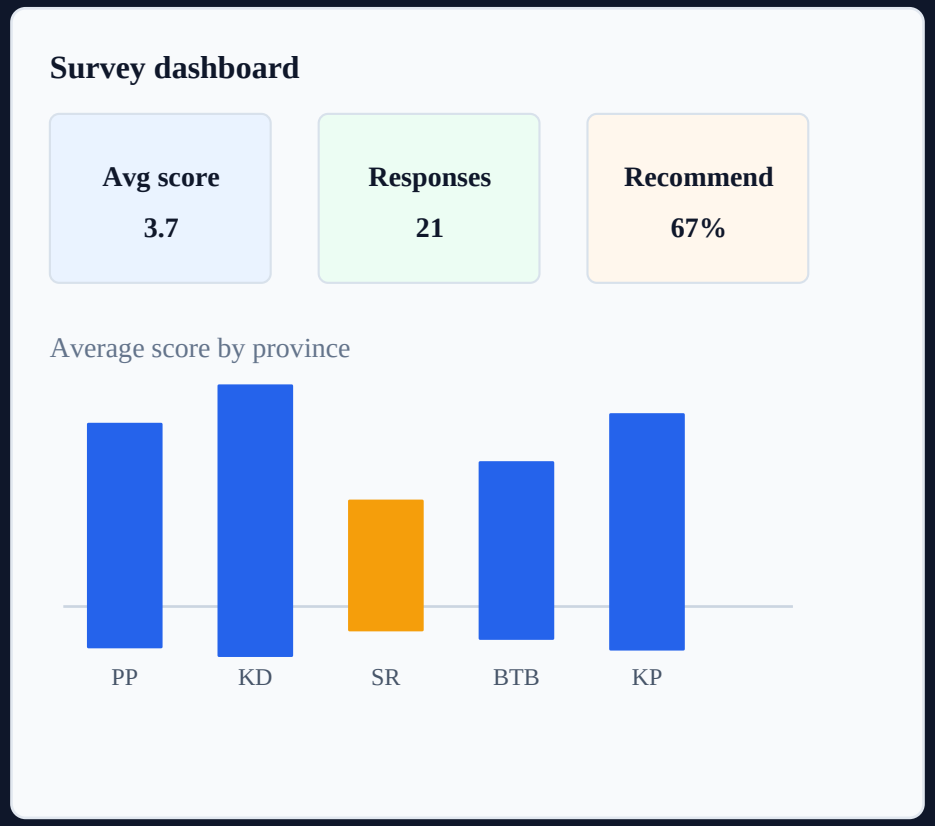
Descriptive Analytics and KPI Tables

Topics for this week:

- descriptive analytics and summary statistics
- grouping and aggregation with groupby
- KPI tables, segmentation, pivot tables, and ranking

Sample project: Online Survey System

Use real survey responses to calculate averages, totals, and KPI tables that help decision makers understand user satisfaction.



From survey responses to KPI insight

clean → group → summarize → compare → explain

Week 3 roadmap

What students should understand and do by the end of this week

1. **Descriptive analytics**
2. **Summary statistics**
3. **groupby and aggregation**
4. **KPI tables**
5. **Pivot tables and ranking**
6. **Business meaning**

Knowledge

Students can explain average, count, min, max, and percentage, and they know why KPI tables matter.

Skills

Students can use describe( ), groupby( ), agg( ), and pivot_table( ) to summarize survey data.

Mindset

Do not stop at numbers. Always ask what the numbers mean for service improvement.

SESSION

Session 1: Descriptive analytics and summary statistics

We start by learning how to describe a dataset before making a deeper argument.

What is descriptive analytics?

Simple idea: use summary numbers to describe what is in the data now or what happened in the past

Input

- Survey responses
- Scores, province, age group, device, recommend, submit time

Work

- Count rows
- Find averages
- Check highest and lowest values
- Compare groups

Value

- See the general picture
- Spot weak areas
- Create simple KPI numbers
- Support next action

Example question for the survey system: “What is the average satisfaction score, and which province or device has the weakest result?”

Common summary statistics for survey data

These small numbers give a quick first picture of quality and performance

Count

How many responses do we have?

21 responses

Mean

What is the average score?

Average score = 3.7

Min / Max

What are the lowest and highest values?

Lowest = 1, Highest = 5

Median

What is the middle score?

Median score = 4

Percentage

What share said yes?

Recommend yes = 67%

Missing values

What information is incomplete?

2 scores missing

First pandas summary commands

Use simple commands to inspect numeric and categorical columns

```
import pandas as pd

df = pd.read_csv("survey_responses.csv")

print(df["score"].describe())
print(df["province"].value_counts())
print(df["recommend"].value_counts(normalize=True))
```

What each command helps us see

score.describe( )

- count, mean, std
- min, 25%, 50%, 75%, max

province.value_counts()

- how many rows per province

recommend.value_counts(normalize =True)

- share or percentage of yes and no

How to explain the output in class

Do not only read numbers. Put them into a sentence.

Example:

“The average survey score is 3.7. Phnom Penh has the most responses. Two thirds of users would recommend the service.”

Grouping and aggregation with groupby

groupby lets us compare segments such as province, age group, and device

```
province_kpi = (
    df.groupby("province")["score"]
      .agg(["count", "mean", "min", "max"])
      .reset_index()
)

print(province_kpi)
```

How groupby works

1. Split rows by one column
2. Apply a summary function
3. Join the result into a small table

This is very useful for province, device, age group, or recommend.

Example output

province	count	mean	min	max
Phnom Penh	4	4.0	3	5
Kandal	3	3.7	3	4
Siem Reap	2	1.5	1	2
Battambang	4	2.0	2	2

Notice the insight:

Siem Reap has the lowest mean score. That makes it a priority segment to investigate.

SESSION

Session 2: KPI tables, pivot tables, and ranking

Now we organize summary results into tables that managers can read quickly.

What is a KPI table?

KPI means a key performance indicator: a number that helps us watch an important goal

A simple KPI table for the survey system

Total responses	**21**
Average score	**3.7**
Recommend ‘yes’ %	**67%**
Mobile share	**57%**
Lowest province	**Siem Reap**

Why KPI tables matter

- fast to read
- easy to compare over time
- good for managers and non-technical users
- can be shown in Flask dashboards
- useful before charts and deeper analysis

Good KPI rule

A KPI should connect to a real question.

Bad: “Many numbers”

Good: “Average score by province”

Bad: “Random table”

Good: “Recommend rate by device”

Segmentation by category or region

Segmentation means we break one big dataset into smaller groups for comparison

By province

Phnom Penh, Kandal, Siem Reap, Battambang

By age group

18-24, 25-34, 35-44, 45-54

By device

mobile, desktop, tablet

By recommend

yes, no

Example business meaning

If mobile users have lower scores than desktop users, the service team should review the mobile experience. If one province reports clearly lower scores, the team can investigate local service quality or response process problems.

Pivot tables and ranking

Pivot tables summarize data by row and column. Ranking helps us order best to worst.

```
pivot = pd.pivot_table(
    df,
    values="score",
    index="province",
    columns="device",
    aggfunc="mean"
)

ranked = province_kpi.sort_values("mean")
```

Why pivot tables help

A pivot table lets us compare two dimensions at the same time.

Example:

rows = province

columns = device

values = average score

Mini pivot view

province	mobile	desktop	tablet
PP	3.8	4.2	3.9
KD	3.5	4.0	3.3
SR	1.5	-	-

Ranking idea: sort the mean score from low to high so weak groups appear first.

Show a KPI table in Flask

Week 3 still uses simple output: pandas builds the table, Flask displays it

```
province_kpi = (df.groupby("province")["score"]
                .agg(["count", "mean"])
                .reset_index())

province_kpi["mean"] = province_kpi["mean"].round(2)

html_table = province_kpi.to_html(index=False)
return render_template("index.html", table=html_table)
```

Province KPI table

province	count	mean
Phnom Penh	4	4.00
Kandal	3	3.67
Siem Reap	2	1.50
Battambang	4	2.00

This is already useful. In Week 4 and later, we can add charts and filters.

Week 3 activities, exit task, and recap

Finish the week with one clear summary and one small result

In-class activity

- Load the survey CSV
- Use describe( ) on score
- Build one groupby table by province
- State one insight in a full sentence

Exit task

- Explain the difference between overall average and segmented average
- Tell the class one useful KPI for the survey system

By the end of Week 3

- Students can describe a dataset
- Students can summarize groups with groupby
- Students can build a simple KPI table for Flask output

Remember: good analytics is not only calculation. It is a clear explanation with useful meaning.

Thanks for attention, Any Question?

Week 3 focus: describe data, build KPI tables, and explain the meaning clearly.